

STEPCLIMBER · REQUIREMENTS

StepClimber requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/stepclimber		
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REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-02	Amish Chadha	Status against SCM-CAL-001 v0.5, recalculated with the approved targets and the 4140 cluster shaft (fatigue assessed); R12 USD 138 over the target after the shaft repricing
0.6	2026-10-02	Amish Chadha	R3 (16 steps/min) and R5 (35 kg) set for the first prototype, and the R7 tilt window tightened to plus or minus 3 degrees for the first loaded trials, by Amish on 2026-10-02 (SCM-DEC-001, items 2 to 4); R3, R5 and R6 met on paper
0.5	2026-10-01	Amish Chadha	Status against SCM-CAL-001 v0.5 for the constructable design (SCM-DDR-003); R3 and R5 now not met; R12 reported against the value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); R3, R5, R7 and R12 revised; status against SCM-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	Record Amish's 2026-09-25 decisions (R1 rating, R11 relaxed to 1,500 mm,

R12 budget kept); status against SCM-CAL-001

0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-24	Amish Chadha	Initial scaffold

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These requirements are checked by calculation in SCM-CAL-001 v0.5, which models the constructable design of SCM-DDR-003 (2026-10-01) on the cluster and drive Amish decided on 2026-09-25 (SCM-DDR-002). On paper, ten are met and four cannot be verified without hardware, after Amish's decisions of 2026-10-02 (SCM-DEC-001): R5 is 35 kg and R3 16 steps/min for the first prototype, and the tilt window is plus or minus 3 degrees for the first loaded trials, which brings R6 within 100 N. Before those decisions R3, R5 and R6 were not met; R12 is reported against the value-engineering target. Targets are still to be revised after user research (see SCM-PRB-001). Decisions are recorded in SCM-DDR-001 to SCM-DDR-003 and indexed in the design decisions register SCM-DEC-001.

The **design load case** is a 60 kg (132 lb) payload on a 34.2 kg truck, 94.2 kg in total, climbing a residential stair with 196 mm (7.75 in) risers and 254 mm (10 in) treads, the steepest the US residential code allows.

Table 1. Requirements and status at TRL 3.

ID	Requirement	Target	Status at TRL 3 (SCM-CAL-001 v0.5)	Verification (later)
R1	Rated stair load	60 kg (132 lb) payload up and down stairs; 100 kg on the flat. Rating decided by Amish, 2026-09-25	Met: safety factors 2.5 (4140 shaft), 1.5 (spider), 3.2 (rail) at a 3 g dropped-step load; shaft fatigue factor 1.8; fatigue of the welded frame not assessed	Proof load
R2	Stair range	Straight flights with risers 100 to 200 mm, treads 250 mm or more and nosing overhang up to 32 mm	Met: lands 66 mm past a square nosing at a 200 mm riser (climbs to 225 mm), needs 240 mm of tread at most, and the arms clear 32 mm overhangs	Stair survey; trial climbs
R3	Climb speed	16 steps/min or more, up and down, at rated load, for the first prototype. Set by Amish on 2026-10-02 (SCM-DEC-001, item 3); revised from 20 to 17 steps/min on 2026-09-25 (SCM-DDR-002)	Met on paper: the climb needs 244 W from the 250 W motor, which gives at most 16.4 steps/min at the 34.2 kg constructable mass (260 W for 17)	Timed climb
R4	Endurance	1,000 or more loaded steps up plus 1,000 down per charge	Met: 1,215	Logged use
R5	Truck mass	35 kg or less with battery for the first prototype; battery removable and 3 kg or less. Set by Amish on 2026-10-02 (SCM-DEC-001, item 2); revised from 25 to 27 kg on 2026-09-25 (SCM-DDR-002)	Met on paper: 34.2 kg (pack 2.6 kg); weighed at TRL 4, with savings tried when parts are bought	Weighing
R6	Operator handle force	100 N or less at the grip while climbing inside the tilt window (R7)	Met on paper with the plus or minus 3 degree window of R7 for the first loaded trials (99 N at the edge; plus or minus 3.1 degrees gives 99 N too); 121 N at a plus or minus 6 degree edge; 76 N at the set angle. Decided by Amish on 2026-10-02 (SCM-DEC-001, item 4)	Force gauge
R7	Tilt control	IMU measures frame angle at 100 Hz or faster; the controller stores the balance angle when a climb starts, shapes motor speed to keep the angle within plus or minus 3 degrees of it for the first loaded trials, and stops and alerts within 0.2 s if the angle leaves that window or the range 15 to 45 degrees back from vertical. The window is widened toward plus or minus 6 degrees only when grip force is measured at 100 N or less, or operators are shown to handle the measured force safely. Set by Amish on 2026-10-02	Not verifiable at TRL 3 (control concept only)	Bench test

ID	Requirement	Target	Status at TRL 3 (SCM-CAL-001 v0.5)	Verification (later)
		(SCM-DEC-001, item 4); tightened from 8 to 6 degrees on 2026-09-25 (SCM-DDR-002)		
R8	Hold on any loss	Load holds on the stair with power off, grip released, battery removed or any fault; drift 5 mm or less in 10 min at rated load	Not verifiable at TRL 3: worm statically self-locking (back-drive efficiency -0.22) and brake margin 7.5 times, but drift needs a test	Hold test
R9	Hold-to-run	Motion only while the dead-man grip is held and a direction is selected; release stops the clusters within 0.2 s	Not verifiable at TRL 3 (circuit concept only)	Bench test
R10	Flat rolling	Push force 40 N or less at rated load on a smooth floor at walking pace, with the clusters parked	Met: 28 N	Pull test
R11	Size	Overall width 600 mm or less; upright height 1,500 mm or less with a fixed handle. Height relaxed from 1,300 mm by Amish, 2026-09-25	Met: 569 mm wide at the wheel faces, 588 mm over the wheel end screws, 1,451 mm tall (model)	Model check; measurement
R12	Affordable	Parts cost for one prototype, charger included, against the value-engineering target of USD 650 (<code>budget_usd</code> , a hypothetical control target set 2026-09-25, SCM-DDR-002)	Estimated cost of the constructable design USD 788: USD 138 over the value-engineering target	Priced BOM (<code>bom/</code> <code>bom.csv</code>)
R13	Battery	24 V class LiFePO4 with BMS and fused output; charging blocked below 0 °C and above 45 °C; full charge in 4 h or less	Met: 3.5 h; protections by specification	Datasheets
R14	Environment	Operate from 0 to 40 °C; electronics IP54 or better; survive rain on outdoor stoops	Not verifiable at TRL 3 (enclosure concept only)	Spray test
R15	Stair and building protection	No steel contact with stairs or walls in normal use; non-marking wheels; skids over stair nosings	Met: the chain case sweeps 53.9 mm on the shaft line against the 54.2 mm that 32 mm nosings leave free; the countershaft guard clears nosings by 80 mm; every frame-fixed outline near the shaft clears every nosing by 10 mm or more	Model check; trial climbs

■ Requirements not met or at risk

- **R5** was not met against 27 kg, by 7.2 kg; it is met against the 35 kg set for the first prototype on 2026-10-02, with 0.8 kg to spare on estimated masses. Making the design buildable named every part: the drive, carried at TRL 3 as a 2.5 kg allowance, weighs 5.5 kg itemised, and the axle plates, chain case, stub axles, uprights, standoffs and fixings add 4.4 kg (SCM-DDR-003). The savings in SCM-DDR-003 item A1 are to be tried when parts are bought.

- **R3** was not met by 4 % at 17 steps/min (260 W needed from a 250 W motor); it is met at the 16 steps/min set for the first prototype on 2026-10-02, needing 244 W from the 250 W motor (a 2 % margin).
- **R6** is met on paper only with the plus or minus 3 degree window set on 2026-10-02 for the first loaded trials; at plus or minus 6 degrees it is 21 % over (121 N). A tight window may cause frequent stops on uneven stairs; it is widened only after grip force is measured.
- **R15 is met with no spare radius:** the chain case is sized to its envelope on the shaft line (10.3 mm from the nearest nosing against 10 mm wanted). A bent case or a stair outside the R2 range could still be struck.
- **R12:** the estimated cost is USD 138 over the USD 650 value-engineering target (the 4140 cluster shaft added USD 12); the main cost drivers are in SCM-DEC-001.
- R7, R8, R9 and R14 depend on control, brake and enclosure behavior that needs hardware to verify.

■ Assumptions

- Payload 60 kg; truck 34.2 kg (mass roll-up in SCM-CAL-001 v0.5); total 94.2 kg.
- The clusters carry the whole weight while lifting (conservative; the operator's grip carries some in practice).
- Center of mass about 500 mm above the cluster shaft; frame tilted about 30 degrees back while climbing.
- The courier stands uphill and pushes or pulls the grip horizontally while climbing.
- Drive efficiencies: motor driver 95 %, brushed motor 80 %, self-locking worm 45 %, first chain stage and bearings 95 %, second chain stage 97 %.
- Rolling resistance coefficient about 0.03 for solid rubber wheels on a smooth floor.
- Stair geometry from the US International Residential Code (196 mm maximum riser, 254 mm minimum tread); national and local codes vary.